



Matrix Theory

14 Projections onto lines

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- Zero dot product $\rightarrow$ orthogonal



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- What about arbitrary angles?

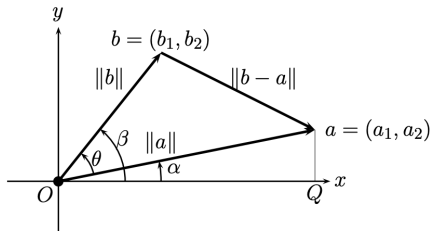


- Zero dot product $\rightarrow$ orthogonal
- What about arbitrary angles?
- Also, how to find the distance from a point to a given subspace?

Inner product and Angle



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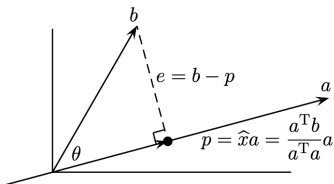


$$\sin(\alpha) = \frac{a_2}{\|a\|}$$

$$\cos(\alpha) = \frac{a_1}{\|a\|}$$

$$\begin{aligned}\cos(\theta) &= \cos(\beta - \alpha) \\ &= \cos(\beta)\cos(\alpha) + \sin(\beta)\sin(\alpha) \\ &= \frac{a_1b_1 + a_2b_2}{\|a\|\|b\|} = \frac{a^T b}{\|a\|\|b\|}\end{aligned}$$

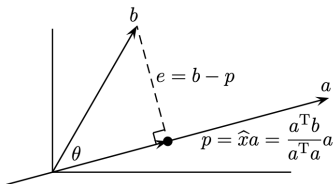
Projection onto a line



Projection p of b onto a

- Must be some multiple of a , i.e., $p = \hat{x}a$

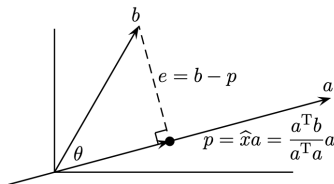
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Projection p of b onto a

- Must be some multiple of a , i.e., $p = \hat{x}a$
- p is the closest point to b on a

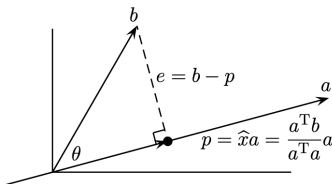
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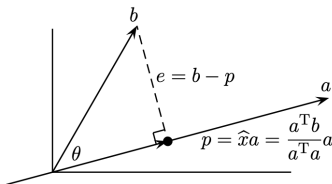
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- Must be some multiple of a , i.e., $p = \hat{x}a$
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- $(b - \hat{x}a) \perp a$, i.e., $a^T(b - \hat{x}a) = 0$

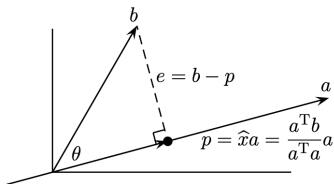
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- $\hat{x} = \frac{a^T b}{a^T a}$ and $p = \hat{x}a = \frac{a^T b}{a^T a}a$

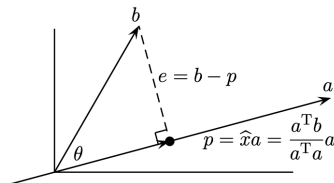
Schwarz inequality



Projection p of b onto a

- Length of e is non-negative

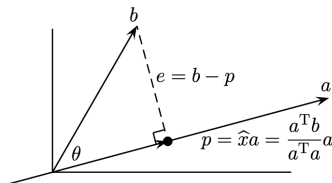
Schwarz inequality



Projection p of b onto a

- Length of e is non-negative
- $\|e\|^2 = \|b - p\|^2 \geq 0$

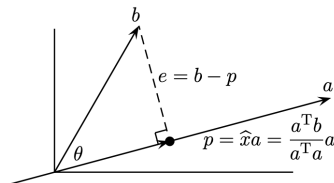
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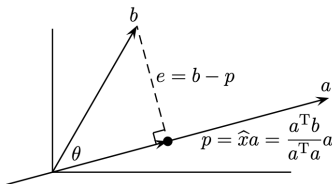
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- $(b^T b)(a^T a) \geq (a^T b)^2$

Schwarz inequality



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Schwarz inequality: $|a^T b| \leq \|a\| \|b\|$

Schwarz inequality



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$$|a^T b| \leq \|a\| \|b\|$$

- Equality holds iff b is a multiple of a

Schwarz inequality



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$$|a^T b| \leq \|a\| \|b\|$$

- Equality holds iff b is a multiple of a
- Also known as the Cauchy-Schwarz-Buniakowsky inequality

Can projection be represented as a matrix/transformation?



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- $p = \hat{x}a$

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- $p = \hat{x}a$
- Let's rewrite: $p = a \frac{a^T b}{a^T a}$

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- $p = \hat{x}a$
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- $p = Pb$, where $P = \frac{aa^T}{a^T a}$

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- $p = Pb$, where $P = \frac{aa^T}{a^T a}$
- P is a rank-1 matrix, symmetric, and $P^2 = P$

Example



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- Project onto a line at θ angle to x-axis



- Projections and Least Squares

Rough Work



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